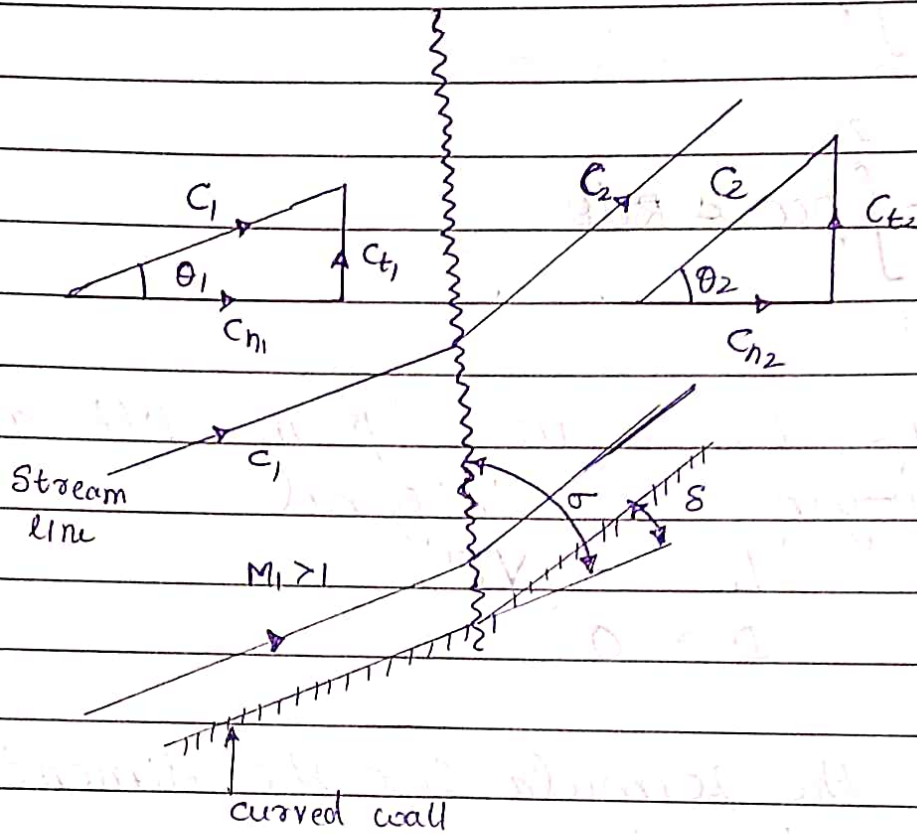
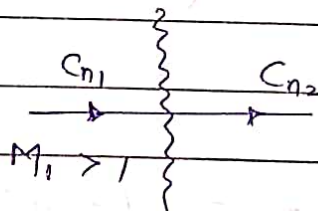


OBLIQUE SHOCK WAVES



(b) Flow through oblique shock



(a) Flow through normal shock

Transformation of a normal shock pattern into oblique

For an oblique shock, the resultant velocity vectors C_1 and C_2 are inclined to the normal direction at angles θ_1 and θ_2 respectively.

$$\tan \theta_1 = \frac{C_{t1}}{C_{n1}} \quad \text{--- (1)}$$

Since there is no pressure gradient along the shock wave

$$C_t = C_{t1} = C_{t2} \quad \text{--- (2)}$$

oblique

For normal shock

$$C_{n2} < C_{n1} \quad \text{--- (3)}$$

From velocity triangle downstream of the shock

$$\tan \theta_2 = \frac{C_{t2}}{C_{n2}} \quad \text{--- (4)}$$

Since $C_{n2} < C_{n1}$ and $C_{t1} = C_{t2}$

$$\tan \theta_2 > \tan \theta_1 \quad \text{--- (5)}$$

Since the flow deviates in anti-clockwise direction after passing through the shock wave. The angle of deviation or deflection is

$$\delta = \theta_2 - \theta_1 \quad \text{--- (6)}$$

From velocity triangle

$$C_1^2 = C_{n1}^2 + C_t^2$$

$$C_2^2 = C_{n2}^2 + C_t^2$$

Since $C_{n2} < C_{n1}$

$$C_2 < C_1$$

Also since $T_2 > T_1$ due to compression

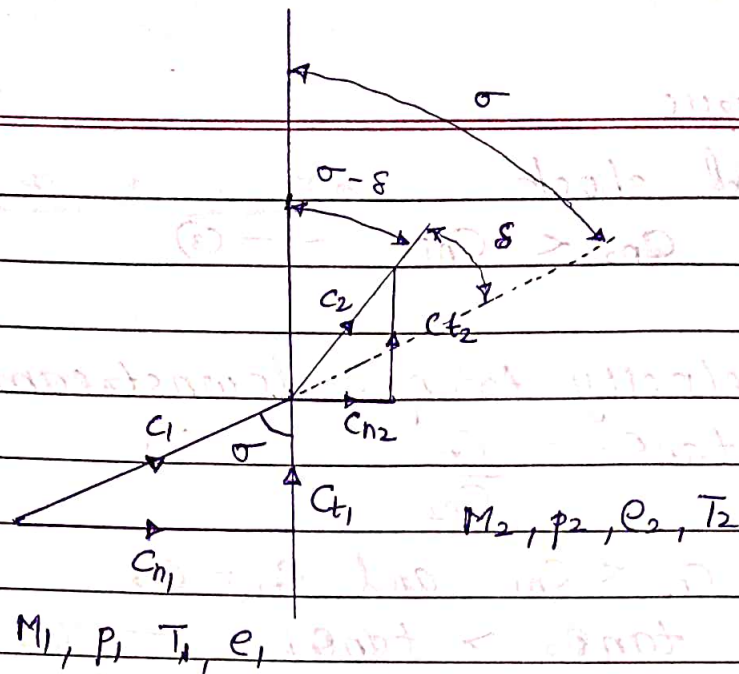
$$\frac{C_2}{\sqrt{\gamma R T_2}} < \frac{C_1}{\sqrt{\gamma R T_1}}$$

$$M_2 < M_1$$

FUNDAMENTAL RELATIONS

Following assumptions are used:

- perfect gas
- steady, adiabatic and frictionless flow through a stationary oblique shock wave,
- absence of external work, and
- absence of body forces.



Continuity equation gives

$$\dot{m} = \rho_1 A C_{n1} = \rho_2 A C_{n2} \quad \text{--- (1)}$$

Area A is perp. to normal vel. component same on both sides

$$\rho_1 C_{n1} = \rho_2 C_{n2} \quad \text{--- (2)}$$

By momentum equation (in tangential direction)

$$\dot{m} C_{t1} = \dot{m} C_{t2}$$

$$C_{t1} = C_{t2} = C_t$$

Momentum equation in normal direction

$$\dot{m} C_{n1} - \dot{m} C_{n2} = (p_2 - p_1) A \quad \text{--- (3)}$$

Substituting (1) in above equation

$$\rho_1 A C_{n1}^2 - \rho_2 A C_{n2}^2 = A (p_2 - p_1)$$

$$\boxed{\rho_1 C_{n1}^2 - \rho_2 C_{n2}^2 = (p_2 - p_1)} \quad \text{--- (4)}$$

Energy equation gives

$$\boxed{h_0 = h_{01} = h_{02} = \text{const.}} \quad \text{--- (5)}$$

$$h_1 + \frac{1}{2} C_1^2 = h_2 + \frac{1}{2} C_2^2$$

$$h_2 - h_1 = \frac{1}{2} (C_1^2 - C_2^2) \quad \text{--- (6)}$$

$$C_p (T_2 - T_1) = \frac{1}{2} (C_1^2 - C_2^2) \quad \text{--- (7)}$$

Considering velocity ~~down~~ upstream and downstream of the ~~shock~~ shock.

$$C_1^2 - C_2^2 = (C_{n1}^2 + C_{t1}^2) - (C_{n2}^2 + C_{t2}^2)$$

$$C_1^2 - C_2^2 = C_{n1}^2 - C_{n2}^2$$

From equation of state

$$h = C_p T = \frac{\gamma}{\gamma-1} R T = \frac{a^2}{\gamma-1} = \frac{\gamma p}{e(\gamma-1)}$$

Substituting the value of h in eq (6)

$$\left[\frac{\gamma}{\gamma-1} \left[\frac{p_2}{e_2} - \frac{p_1}{e_1} \right] \right] = \frac{1}{2} (C_{n1}^2 - C_{n2}^2) \quad \text{--- (8)}$$

Stagnation value can be written as

$$h_0 = C_p T_0 = \frac{\gamma}{\gamma-1} R T_0 = \frac{\gamma}{\gamma-1} \frac{p_0}{e_0} = \frac{a_0^2}{\gamma-1}$$

$$h_0 = \frac{a_0^2}{\gamma-1} = \frac{1}{2} \left(\frac{\gamma+1}{\gamma-1} \right) a^{*2}$$

$$h_0 = \frac{\gamma}{\gamma-1} \frac{p_0}{e_0} = \frac{a_0^2}{\gamma-1} = \frac{1}{2} \frac{\gamma+1}{\gamma-1} a^{*2}$$

PRANDTL'S EQUATION

From energy equation

$$\frac{\gamma}{\gamma-1} \left[\frac{p_2}{e_2} - \frac{p_1}{e_1} \right] = \frac{1}{2} (C_{n1}^2 + C_{t1}^2) - \frac{1}{2} (C_{n2}^2 + C_{t2}^2)$$

On rearrangement

$$\frac{\gamma}{\gamma-1} \frac{p_1}{\rho_1} + \frac{1}{2} (C_{n1}^2 + C_{t1}^2) = \frac{\gamma}{\gamma-1} \frac{p_2}{\rho_2} + \frac{1}{2} (C_{n2}^2 + C_{t2}^2)$$

$$\frac{\gamma}{\gamma-1} \frac{p_1}{\rho_1} + \frac{1}{2} (C_{n1}^2 + C_{t1}^2) = \frac{1}{2} \frac{\gamma+1}{\gamma-1} a^{*2}$$

(considering Upstream only--)

From above equation keeping p_1 only on L.H.S.

$$p_1 = \frac{\gamma-1}{2\gamma} \rho_1 \left\{ \frac{\gamma+1}{\gamma-1} a^{*2} - (C_{n1}^2 + C_{t1}^2) \right\}$$

$$\text{or } p_1 = \rho_1 \left\{ \frac{\gamma+1}{2\gamma} a^{*2} - \frac{\gamma-1}{2\gamma} (C_{n1}^2 + C_{t1}^2) \right\}$$

Similarly we can write an equation for p_2 considering downstream of shock

$$p_2 = \rho_2 \left\{ \frac{\gamma+1}{2\gamma} a^{*2} - \frac{\gamma-1}{2\gamma} (C_{n2}^2 + C_{t2}^2) \right\}$$

Substituting the value of p_1 and p_2 in eqn)

$$\begin{aligned} \rho_1 C_{n1}^2 - \rho_2 C_{n2}^2 &= p_2 - p_1 \\ &= \rho_2 \left\{ \frac{\gamma+1}{2\gamma} a^{*2} - \frac{\gamma-1}{2\gamma} (C_{n2}^2 + C_{t2}^2) \right\} \\ &\quad - \rho_1 \left\{ \frac{\gamma+1}{2\gamma} a^{*2} - \frac{\gamma-1}{2\gamma} (C_{n1}^2 + C_{t1}^2) \right\} \\ &= \frac{\gamma+1}{2\gamma} a^{*2} (\rho_2 - \rho_1) + \frac{\gamma-1}{2\gamma} \{ \rho_1 (C_{n1}^2 + C_{t1}^2) - \rho_2 (C_{n2}^2 + C_{t2}^2) \} \end{aligned}$$

Using $C_{t1} = C_{t2} = C_t$ and rearrangement

$$\begin{aligned} \rho_1 C_{n1}^2 - \rho_2 C_{n2}^2 &= \frac{\gamma+1}{2\gamma} a^{*2} (\rho_2 - \rho_1) \\ &\quad + \frac{\gamma-1}{2\gamma} (\rho_1 C_{n1}^2 - \rho_2 C_{n2}^2 + C_t^2 (\rho_1 - \rho_2)) \end{aligned}$$

$$\frac{\gamma+1}{2\gamma} a^{*2} (e_2 - e_1) = (e_1 c_{n1}^2 - e_2 c_{n2}^2) - \frac{\gamma-1}{2\gamma} (e_1 c_{n1}^2 - e_2 c_{n2}^2) - \left(\frac{\gamma-1}{2\gamma}\right) c_t^2 (e_1 - e_2)$$

$$\frac{\gamma+1}{2\gamma} a^{*2} (e_2 - e_1) = (e_1 c_{n1}^2 - e_2 c_{n2}^2) \left[1 - \frac{\gamma-1}{2\gamma}\right] - \frac{\gamma-1}{2\gamma} c_t^2 (e_1 - e_2)$$

$$\frac{\gamma+1}{2\gamma} a^{*2} (e_2 - e_1) = \frac{\gamma+1}{2\gamma} (e_1 c_{n1}^2 - e_2 c_{n2}^2) - \frac{\gamma-1}{2\gamma} c_t^2 (e_1 - e_2)$$

$$a^{*2} = \frac{e_1 c_{n1}^2 - e_2 c_{n2}^2}{(e_2 - e_1)} + \frac{\gamma-1}{\gamma+1} c_t^2$$

$$a^{*2} - \frac{\gamma-1}{\gamma+1} c_t^2 = \frac{c_{n1} c_{n2}}{e_2 - e_1} \left(e_1 \frac{c_{n1}}{c_{n2}} - e_2 \frac{c_{n2}}{c_{n1}} \right)$$

But from (2) we know $\frac{c_{n1}}{c_{n2}} = \frac{e_2}{e_1}$

$$\therefore \boxed{a^{*2} - \frac{\gamma-1}{\gamma+1} c_t^2 = c_{n1} c_{n2}}$$

This is Prandtl's equation for oblique shocks

RANKINE HUGONIOT EQUATION

Equation (4) can be written as

$$p_2 - p_1 = e_1 c_{n1}^2 \left[1 - \frac{e_2 c_{n2}^2}{e_1 c_{n1}^2} \right]$$

$$\text{We know } \frac{c_{n1}}{c_{n2}} = \frac{e_2}{e_1} \therefore \frac{c_{n2}^2}{c_{n1}^2} = \frac{e_1^2}{e_2^2}$$

On substituting

$$p_2 - p_1 = e_1 c_{n_1}^2 \left(1 - \frac{e_1}{e_2}\right) = (e_2 - e_1) \frac{e_1 c_{n_1}^2}{e_2}$$

$$c_{n_1}^2 = \frac{p_2 - p_1}{e_2 - e_1} \frac{e_2}{e_1}$$

$$p_2 - p_1 = e_2 c_{n_2}^2 \left[\frac{e_1 c_{n_1}^2}{e_2 c_{n_2}^2} - 1 \right]$$

$$p_2 - p_1 = e_2 c_{n_2}^2 \left[\frac{e_1}{e_2} \left(\frac{e_2}{e_1}\right)^2 - 1 \right] = e_2 c_{n_2}^2 \left[\frac{e_2 - e_1}{e_1} \right]$$

$$p_2 - p_1 = \frac{e_2 c_{n_2}^2 (e_2 - e_1)}{e_1}$$

$$c_{n_2}^2 = \frac{p_2 - p_1}{e_2 - e_1} \frac{e_1}{e_2}$$

Substituting the value of c_{n_1} and c_{n_2} in (8)

$$\frac{\gamma}{\gamma-1} \left(\frac{p_2}{e_1} - \frac{p_1}{e_1} \right) = \frac{1}{2} (c_{n_1}^2 - c_{n_2}^2)$$

$$\begin{aligned} \frac{\gamma}{\gamma-1} \left(\frac{p_2}{e_1} - \frac{p_1}{e_1} \right) &= \frac{1}{2} \left[\frac{p_2 - p_1}{e_2 - e_1} \frac{e_2}{e_1} - \frac{p_2 - p_1}{e_2 - e_1} \frac{e_1}{e_2} \right] \\ &= \frac{1}{2} \frac{(p_2 - p_1)(e_2 + e_1)}{e_1 e_2} \end{aligned}$$

On rearranging

$$\begin{aligned} 2\gamma (e_1 p_2 - e_2 p_1) &= (\gamma-1)(p_2 - p_1)(e_2 + e_1) \\ p_2 \{ 2\gamma e_1 - (\gamma-1)(e_2 + e_1) \} &= p_1 \{ 2\gamma e_2 - (\gamma-1)(e_2 + e_1) \} \end{aligned}$$

$$\frac{p_2}{p_1} = \frac{2\gamma e_2 - (\gamma-1)(e_2 + e_1)}{2\gamma e_1 - (\gamma-1)(e_2 + e_1)}$$

$$p_2 \{ (\gamma-1)e_2 - (\gamma+1)e_1 \} = p_1 \{ (\gamma-1)e_1 - (\gamma+1)e_2 \}$$

$$\frac{p_2}{p_1} = \frac{(\gamma-1)e_1 - (\gamma+1)e_2}{(\gamma-1)e_2 - (\gamma+1)e_1}$$

$$\boxed{\frac{p_2}{p_1} = \frac{\gamma+1}{\gamma-1} \frac{e_2}{e_1} - 1}$$

$$\frac{\gamma+1}{\gamma-1} - \frac{e_2}{e_1}$$

Rearranging $2\gamma(e_1 p_2 - e_2 p_1) = (\gamma-1)(p_2 - p_1)(e_2 + e_1)$
by taking e_2 and e_1 as common factors
and taking ratios

$$\frac{e_2}{e_1} = \frac{(\gamma+1)p_2 + (\gamma-1)p_1}{(\gamma+1)p_1 + (\gamma-1)p_2}$$

$$\boxed{\frac{e_2}{e_1} = \frac{\frac{\gamma+1}{\gamma-1} \frac{p_2}{p_1} + 1}{\frac{\gamma+1}{\gamma-1} + \frac{p_2}{p_1}}}$$

We have already framed an equation for C_{n1}^2 and C_{n2}^2

$$\therefore C_{n1}^2 C_{n2}^2 = \left(\frac{p_2 - p_1}{e_2 - e_1} \right)^2$$

From Prandtl's equation

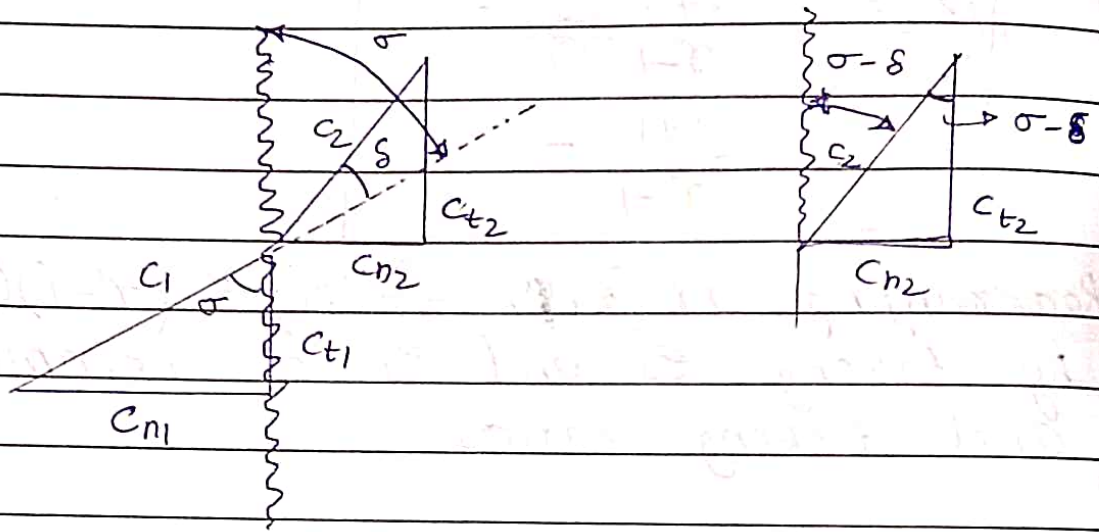
$$a^{*2} - \frac{\gamma-1}{\gamma+1} C_t^2 = C_{n1} C_{n2}$$

$$\boxed{a^{*2} - \frac{\gamma-1}{\gamma+1} C_t^2 = \frac{p_2 - p_1}{e_2 - e_1}}$$

TRIGONOMETRICAL RELATIONS

From the velocity triangle

$$C_{n1} = C_1 \sin \sigma$$



$$C_{t1} = C_1 \cos \sigma$$

$$C_{n2} = C_2 \sin (\sigma - \delta)$$

$$C_{t2} = C_2 \cos (\sigma - \delta)$$

Since $C_{t1} = C_{t2}$

$$C_1 \cos \sigma = C_2 \cos (\sigma - \delta)$$

$$\frac{C_1}{C_2} = \frac{\cos (\sigma - \delta)}{\cos \sigma} \quad \text{--- (1)}$$

We know $\rho_1 C_{n1} = \rho_2 C_{n2}$ (continuity eq-)

$$\rho_1 C_1 \sin \sigma = \rho_2 C_2 \sin (\sigma - \delta)$$

$$\frac{\rho_2}{\rho_1} = \frac{C_1 \sin \sigma}{C_2 \sin (\sigma - \delta)} \quad \text{--- (2)}$$

Substituting the value of C_1/C_2 from (1) in (2)

$$\frac{\rho_2}{\rho_1} = \frac{\cos (\sigma - \delta)}{\cos \sigma} \left[\frac{\sin \sigma}{\sin (\sigma - \delta)} \right]$$

$$\frac{\rho_2}{\rho_1} = \frac{\tan \sigma}{\tan (\sigma - \delta)}$$

We know $p_2 - p_1 = e_1 c_{n1}^2 - e_2 c_{n2}^2$

$$p_2 - p_1 = e_1 c_{n1}^2 \left[1 - \frac{e_2 c_{n2}^2}{e_1 c_{n1}^2} \right]$$

$$p_2 - p_1 = e_1 c_1^2 \sin^2 \sigma \left[1 - \frac{c_{n1} c_{n2}^2}{c_{n2} c_{n1}^2} \right]$$

$$p_2 - p_1 = e_1 c_1^2 \sin^2 \sigma \left[1 - \frac{e_2}{e_1} \right]$$

$$M_1^2 = \frac{c_1^2}{a_1^2} \quad \text{or} \quad Q^2 = M_1^2 a_1^2 = M_1^2 \gamma \frac{p_1}{e_1}$$

$$e_1 c_1^2 = \gamma p_1 M_1^2$$

Therefore $p_2 - p_1 = \gamma p_1 M_1^2 \sin^2 \sigma \left[1 - \frac{e_2}{e_1} \right]$

$$\frac{p_2}{p_1} - 1 = \gamma M_1^2 \sin^2 \sigma \left[1 - \frac{e_2}{e_1} \right]$$

$$\text{or} \quad \left[\frac{p_2}{p_1} = 1 + \gamma M_1^2 \sin^2 \sigma \left[1 - \frac{e_2}{e_1} \right] \right] \quad \text{--- (3)}$$

In a similar manner taking $e_2 c_{n2}^2$ from eq $p_2 - p_1 = e_1 c_{n1}^2 - e_2 c_{n2}^2$ we can deduce a pressure ratio relation as

$$\left[\frac{p_1}{p_2} = 1 + \gamma M_2^2 \sin^2(\sigma - \delta) \left[1 - \frac{e_2}{e_1} \right] \right] \quad \text{--- (4)}$$

DENSITY RATIO

From Rankine-Hugoniot Equation, we have pressure ratio

$$\frac{p_2}{p_1} = \frac{\gamma + 1}{\gamma - 1} \frac{e_2 - 1}{e_1}$$

$$\frac{\gamma + 1}{\gamma - 1} \frac{e_2}{e_1}$$

Equating this equation (3) we get

$$\frac{\frac{\gamma+1}{\gamma-1} \frac{e_2}{e_1} - 1}{\frac{\gamma+1}{\gamma-1} - \frac{e_2}{e_1}} = 1 + \gamma M_1^2 \sin^2 \sigma \left[1 - \frac{e_1}{e_2} \right]$$

On rearranging and taking $\frac{e_2}{e_1}$ on to L.H.S.

$$\frac{e_2}{e_1} = \frac{(\gamma+1) M_1^2 \sin^2 \sigma}{2 + (\gamma-1) M_1^2 \sin^2 \sigma}$$

Pressure Ratio

Substituting e_1/e_2 in eq (3)

$$\frac{p_2}{p_1} = 1 + \gamma M_1^2 \sin^2 \sigma \left\{ 1 - \frac{2 + (\gamma-1) M_1^2 \sin^2 \sigma}{(\gamma+1) M_1^2 \sin^2 \sigma} \right\}$$

On simplification,

$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma+1} (M_1^2 \sin^2 \sigma - 1)$$

$$\frac{p_2}{p_1} = \frac{2\gamma}{\gamma+1} M_1^2 \sin^2 \sigma - \frac{\gamma-1}{\gamma+1}$$

Temperature Ratio

$$\frac{T_2}{T_1} = \frac{(p_2/p_1)}{(e_2/e_1)}$$

On substitution

$$\frac{T_2}{T_1} = \frac{[2\gamma M_1^2 \sin^2 \sigma - (\gamma-1)] [2 + (\gamma-1) M_1^2 \sin^2 \sigma]}{(\gamma+1)^2 M_1^2 \sin^2 \sigma}$$

Stagnation Pressure Ratio

$$\frac{p_{01}}{p_1} = \left(\frac{T_{01}}{T_1} \right)^{\gamma/\gamma-1}$$

$$\frac{p_{02}}{p_2} = \left(\frac{T_{02}}{T_2} \right)^{\gamma/\gamma-1}$$

$$\frac{p_{02}}{p_{01}} = \left[\frac{T_{02}}{T_2} \times \frac{T_1}{T_{01}} \right]^{\gamma/\gamma-1} \frac{p_2}{p_1}$$

Since $T_{02} = T_{01}$

$$\frac{p_{02}}{p_{01}} = \left(\frac{T_1}{T_2} \right)^{\gamma/\gamma-1} \frac{p_2}{p_1}$$

On substitution

$$\frac{p_{02}}{p_{01}} = \left[\frac{(\gamma+1) M_1^2 \sin^2 \sigma}{(\gamma-1) M_1^2 \sin^2 \sigma + 2} \right]^{\gamma/\gamma-1} \left[\frac{\gamma+1}{2\gamma M_1^2 \sin^2 \sigma - (\gamma-1)} \right]^{\gamma/\gamma-1}$$

Change of Entropy

$$\frac{s_2 - s_1}{R} = \frac{\Delta s}{R} = - \ln \frac{p_{02}}{p_{01}}$$

On substitution

$$\frac{\Delta s}{R} = \frac{\gamma}{\gamma-1} \ln \left[\frac{(\gamma-1) M_1^2 \sin^2 \sigma + 2}{(\gamma+1) M_1^2 \sin^2 \sigma} \right] + \frac{1}{\gamma-1} \ln \left[\frac{2\gamma M_1^2 \sin^2 \sigma - (\gamma-1)}{(\gamma+1)} \right]$$

Deflection Angle

$$\tan \sigma = \frac{M_1^2 \sin 2\sigma - 2 \cot \sigma}{2 + M_1^2 (\gamma + \cot^2 \sigma)}$$

OBLIQUE SHOCK (PROBLEMS)

Air approaches a symmetrical wedge ($\delta = 15^\circ$) at a Mach no. of 2. Determine for the strong and weak waves (a) wave angle (b) pressure ratio (c) density ratio, (d) temperature ratio and (e) downstream Mach number

$$\tan \delta = 2 \cot \sigma \frac{M_1^2 \sin^2 \sigma - 1}{2 + \gamma M_1^2 + M_1^2 (1 - 2 \sin^2 \sigma)}$$

For $\delta = 15^\circ$ and $M_1 = 2$ we get

$$\tan 15 = 2 \cot \sigma \frac{2^2 \sin^2 \sigma - 1}{2 + 1.4 \times 2^2 + 2^2 (1 - 2 \sin^2 \sigma)}$$

Solving for σ

$$0.1339 = \frac{4 \sin^2 \sigma - 1}{11.6 - 8 \sin^2 \sigma} \cot \sigma \quad \text{--- (A)}$$

(a) $\sigma = 79.8^\circ$

(b) Pressure ratio : $\frac{p_2}{p_1} = \frac{2\gamma M_1^2 \sin^2 \sigma}{\gamma + 1} - \left(\frac{\gamma - 1}{\gamma + 1} \right)$

$$\frac{p_2}{p_1} = \frac{2 \times 1.4}{1.4 + 1} \times 2^2 \sin^2 79.8 - \left(\frac{1.4 - 1}{1.4 + 1} \right)$$

$$\frac{p_2}{p_1} = 4.353$$

(c) $\sigma - \delta = 79.8 - 15 = 64.8$

$$\frac{e_2}{e_1} = \frac{\tan \sigma}{\tan(\sigma - \delta)} = \frac{\tan 79.8}{\tan 64.8} = 2.615$$

$$\therefore \frac{e_2}{e_1} = 2.615$$

(d) $\frac{T_2}{T_1} = \frac{p_2/p_1}{e_2/e_1} = \frac{4.353}{2.615} = 1.665$

$$\text{we know } \frac{p_1}{p_2} = 1 + \gamma M_2^2 \sin^2(\sigma - \delta) \left[1 - \frac{e_2}{e_1} \right]$$

$$\Rightarrow \frac{1}{4.353} = 1 + 1.4 M_2^2 \sin^2(79.8 - 15) \left[1 - 2.191 \right]$$

On simplification $M_2 = 0.645$

For value of $\sigma < 90$, on solving eq (A), we get $\sigma_{weak} = 45.3$

$$\therefore \frac{p_2}{p_1} = \frac{2\gamma}{\gamma+1} M_1^2 \sin^2 \sigma - \frac{\gamma-1}{\gamma+1}$$

$$\frac{p_2}{p_1} = \frac{2 \times 1.4}{1.4+1} \times 2^2 \sin^2 45.3 - \frac{1.4-1}{1.4+1} = 2.191$$

$$(i) \quad \sigma - \delta = 45.3 - 15 = 30.3^\circ$$

$$(ii) \quad \frac{e_2}{e_1} = \frac{\tan \sigma}{\tan(\sigma - \delta)} = \frac{\tan 45.3}{\tan 30.3} = 1.729$$

$$(iii) \quad \frac{T_2}{T_1} = \frac{(p_2/p_1)}{(e_2/e_1)} = \frac{2.191}{1.729} = 1.267$$

$$(iv) \quad \frac{p_1}{p_2} = 1 + \gamma M_2^2 \sin^2(\sigma - \delta) \left[1 - \frac{e_2}{e_1} \right]$$

$$\therefore \frac{1}{2.191} = 1 + 1.4 \times M_2^2 \sin^2(30.3) \times \left[1 - 1.729 \right]$$

$$M_2 = 1.448$$